Transactions of the Royal Society of Edinburgh: Earth Sciences, **73**, 59–65, 1982

Evidence for metamorphic remagnetisation of upper Precambrian tillite in the Dalradian Supergroup of Scotland

M. Stupavsky, D. T. A. Symons and C. P. Gravenor

ABSTRACT: A conglomerate test using 36 clast cores (76 specimens) from the upper Precambrian Port Askaid Tillite of the Dalradian Supergroup in the Garvellach–Islay region of Scotland shows that the tillite was remagnetised during orogenic activity in Lower Ordovician time. After 550°C thermal demagnetisation, the clast remanence directions give a mean of 155°, 14° (α_{95}–12°). This is similar to the mean from four tillite matrix sites of 162°, 4° (α_{95}–9°) and the mean 'primary component direction' of 166°, −12° (α_{95}–6°) previously determined from the interbedded siltstones. This indicates that the matrix and siltstones were also remagnetised. These mean directions give virtual geomagnetic poles after tilt correction that fall within the cluster of British Ordovician poles from rocks in the Caledonian orogen, and thereby indicate a prefolding Ordovician age for the remagnetisation. Thermal demagnetisation studies and observations from polished and thin sections of tillite clast and matrix specimens show that secondary hematite derived from partially to completely altered magnetite is the dominant carrier of the metamorphic remagnetisation. These results show that remanence data from the Port Askaid Tillite cannot be used to support the hypothesis that a worldwide glaciation occurred during the Late Precambrian.

KEY WORDS: Caledonian orogen, clast, hematite, matrix, palaeomagnetism, Port Askaid Tillite, remanence data, worldwide glaciation.

Palaeomagnetic studies of upper Precambrian glacial deposits from various parts of the world have yielded equatorial palaeolatitudes (Harland & Bidgood 1959; Tarling 1974; Morris 1977; Kröner *et al.* 1980; McWilliams & McElhinny 1980). Two interpretations of these results have been proposed. The first favours the concept that the isolated remanences giving rise to these low palaeolatitudes are primary magnetisations acquired shortly after deposition of the till. This interpretation suggests a worldwide Late Precambrian glaciation with the glaciers extending to the equator (e.g. Harland 1964). The second interpretation favours the concept that the isolated remanences are secondary remagnetisations acquired far later during metamorphism of the lithified tillites. This latter interpretation suggests that the tills were deposited at high palaeolatitudes under conditions similar to those that prevailed in the Northern Hemisphere during the Pleistocene, and that after lithification to tillites they were carried by continential drift to low palaeolatitudes where they were remagnetised by metamorphism. The presence of metamorphic remagnetisation in tillites may be detected by the palaeomagnetic conglomerate test (Graham 1949) using the larger tillite clasts. Significant remagnetisation is demonstrated if the clast remanence directions form a coherent cluster and especially if the cluster mean direction is statistically the same as the mean direction from the tillite matrix. This paper reports on the results of a palaeomagnetic conglomerate test on the upper Precambrian tillite exposed in the Garvellach–Islay region of Scotland.

1. Geology and previous study

The Port Askaid Tillite is the basal formation of the Middle Dalradian Argyll Group of the Dalradian Supergroup (Harris & Pitcher 1975). It crops out as a well defined unit along most of the Caledonian orogenic belt in the Metamorphic Caledonides (Watson 1975). It has been tentatively correlated with other upper Precambrian tillites in Spitzbergen (Kulling 1934), in eastern Greenland (Schaub 1950), and in northern Scandanavia (Spjeldnaes 1964) where the Varanger Tillite has been radiometrically dated at 668 ± 23 m.y. (Pringle 1972). The 669 ± 17 m.y. date on the post-tillite Portsoy granite also sets a lower limit on the tillite deposition (Pankhurst 1974). In the Garvellach–Islay region, the Port Askaid Tillite is >700 m thick, and it conformably overlies Lower Dalradian marine limestones of the Appin Group (Spencer 1971). Lower Cambrian fossils first appear in overyling limestones and shales of the Middle Dalradian (Harris & Pitcher 1975). Dalradian sedimentation terminated prior to the onset of metamorphism at 520 ± 30 m.y. (Leggo & Pidgeon 1970) and the emplacement of intrusive masses in Late Cambrian time (Pidgeon 1969; Pankhurst 1970). Regional metamorphism reached greenschist and amphibolite facies levels. The climax of metamorphism and deformation occurred from about 490 to 510 m.y. at the Upper Cambrian–Lower Ordovician boundary with uplift–cooling ages ranging down to about 430 to 450 m.y. in late early Ordovician to early Silurian time (Dewey & Pankhurst 1970; Moorbath 1975). Deformation during this orogenic event tectonically tilted the Port Askaid Tillite to give dips of ~30°S.

Tarling (1974) and Abouzakhm and Tarling (1975) did palaeomagnetic studies of the Port Askaid Tillite using exposures on the Garvellach Islands. Tarling (1974) sampled siltstones at eight sites and tillite matrix at one site. He isolated two stable characteristic magnetisations in the silstones. One component at 59°, −57° (declination, inclination) (α_{95}–10°) was interpreted as a metamorphic remagnetisation acquired in Ordovician to Silurian time. The second

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component at 166°, −12° (α_{95}–6°) was tentatively interpreted to be the primary magnetisation acquired during deposition when the unit was at ~10°S palaeolatitude. This interpretation supports the concept of a worldwide glaciation in the Late Precambrian. It should be noted, however, that Tarling (1974, p. 177) states: "I do not consider the low palaeolatitude for these Scottish tillites to have been conclusively proved as their age of magnetisation has not been established equivocally, . . ." The study of Abouzakhm and Tarling (1975) examined the magnetic fabric and thermal demagnetisation characteristics of the siltstones. This work led to the conclusion that the rocks had not been metamorphosed sufficiently either dynamically or thermally to affect the remanence carriers. The objective of this present study is to use the conglomerate test to see if the 166°, −12° component may be a later metamorphic event, and to use polished and thin section observations to see if chemical alteration may have caused new remanence carriers to have been formed.

2. Sampling and measurement

Tillite clasts, matrix and associated siltstones were sampled on Garbh Eileach island (Fig. 1). The tillite clasts and matrix were drilled from a shoreline outcrop about 200 m E of Tarling's (1974) sites 1 to 8. A total of 36 cores of ~10 cm length were drilled in clasts for the conglomerate test, and 5 cores were drilled at each of four matrix sites (7, 9, 10, 11). These four matrix sites span a 12 m vertical thickness of the tillite with ~4 m of separation between the sites. Tillite matrix site 12 was drilled ~100 m W of Tarling's site 9. A further six siltstone sites were drilled in the same outcrops used by Tarling (1974). All cores were drilled about vertically and oriented with a magnetic compass. No significant magnetic deviations were detected in the ~25% of the cores for which sun compass orientations were also obtained.

Two or three 2·4 cm long specimens were cut from each core. The specimens were stored in a magnetically shielded room for. more than one week before the remanence measurements were made to allow the most recently acquired viscous remanent magnetisation components to decay away. All the remanent magnetisation measurements were carried out on a Schonstedt SSM-1A spinner magnetometer using either a 3- or 6-spin system of measurement. The within-specimen remanence angular standard deviation (ASD) was also computed from the two (3-spin) and four (6-spin) redundant measurements of each Cartesian component of the remanent vector. The ASD gives a measure of the within-specimen remanence homogeneity and it was used as the screening criterion for selecting only homogeneously (reliably) magnetised specimens for interpretation (Harrison 1980; Lowrie *et al.* 1980; Briden & Arthur 1981). Alternating field (AF) and thermal demagnetisation of the specimens were carried out on Schonstedt GSD-1 alternating field and TSD-1 thermal demagnetisers.

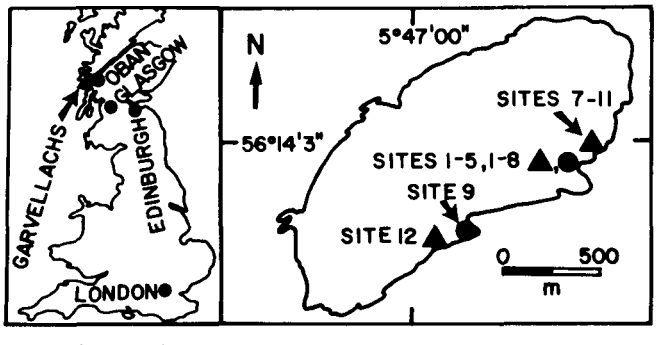


Figure 1 Location map of Garbh Eileach island showing the sampling sites of this study (triangles) and of Tarling (1974) (circles).

3. Remanence results

3.1. Tillite matrix

The tillite matrix at three sites (7, 9, 10) of the five sites exhibits stable magnetisation with a relatively high natural remanent magnetisation (NRM) intensity (mean $1{\cdot}7 \times 10^{-1}$ Am^{-1}; s.d. $1{\cdot}0 \times 10^{-1}$ Am^{-1}). The NRM has good within-specimen (mean ASD 6·3°, s.d. 2·4°) and within-site (mean α_{95} 8·9°, s.d. 2·4°) remanence homogeneity (Table 1). During step AF and thermal demagnetisation of their pilot specimens, the remanence directions move to a stable end point (Fig. 2, spec. 7, 9, 10). The remanence intensity on AF demagnetisation drops rapidly to ~20 mT and thereafter is virtually unchanged up to 100 mT at from ~40 to 70% of the NRM intensity. This indicates minor less stable magnetite and abundant more stable hematite remanence carriers (Fig. 2). Similarly, the thermal intensity decay curves show a dominance of hematite carriers. The mean remanence direction based on the three homogeneously magnetised site means also shows progressive movement from the NRM to the stable end point at 670°C on step demagnetisation (Table 2). The best defined tilt-corrected (mean strike N 95°E and dip 31°S) mean direction is 162°, 4° (α_{95} − 9°) after 550°C demagnetisation which is close to

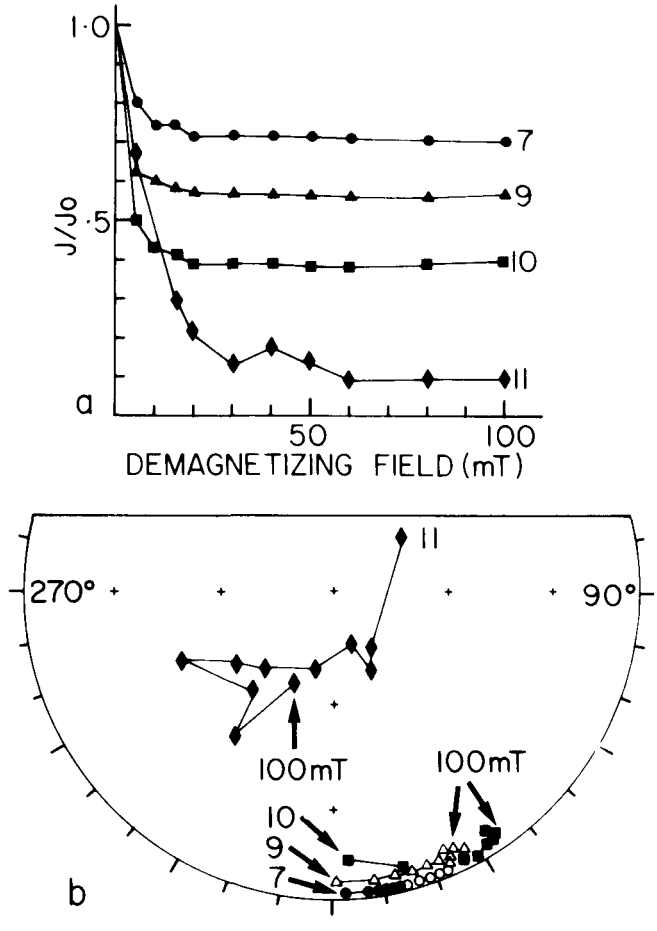


Figure 2 AF step demagnetisation of 4 tillite matrix pilot specimens showing (a) the ratio of the measured remanence to the initial NRM remanence intensity (J/Jo) as a function of the peak AF demagnetizing field in milliTesla (mT) and (b) the remanence directions plotted on a Schmidt equal area stereonet with solid (open) indicating down (up) directions; the specimen number is by the NRM direction.

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Table 1 Tilt corrected tillite matrix remanence data

Site number[1]	Number of specimens[2]	Treatment	Declination (degrees)	Inclination (degrees)	Length of vector resultant, R^3	α_{95}[3] (degrees)	ASD[4] (degrees)
7	15	NRM	181·8	0·2	14·6110	6·5	all specimens
9	15		182·2	10·7	14·6116	6·5	,,
10	12		176·2	14·4	11·8884	4·4	,,
11	15		151·1	56·3	11·5748	21·7	,,
12	12		185·8	63·2	9·1643	25·4	,,
MEAN	3[5]		180·1	8·4	2·9802	12·3	
7	13	20MT	170·9	−4·9	12·2800	10·5	≤10
9	12	,,	163·2	0·2	11·7814	6·2	,,
10	11	,,	156·6	5·7	10·5378	10·0	,,
11	13	,,	168·6	35·5	10·6125	20·6	≤20
12	9	,,	166·7	22·5	6·4376	35·0	,,
MEAN	3[5]		163·6	0·3	2·9759	13·6	
7	12	550°C	167·1	6·8	11·4178	10·3	≤10
9	14	,,	164·8	0·8	12·5910	13·8	,,
10	11	,,	161·7	−1·7	10·7325	7·6	,,
11	8	,,	152·6	9·2	7·6776	12·2	,,
12	10	,,	180·1	47·5	8·1785	24·2	≤20
MEAN	4[5]		161·6	3·8	3·9700	9·2	
7	12	650°C	157·1	−6·1	11·0034	13·7	≤10
9	13	,,	139·5	−5·1	11·2027	17·3	,,
10	10	,,	152·4	−12·3	9·1730	15·3	,,
11	14	,,	155·9	1·8	3·7071	73·7	≤20
12	10	,,	160·8	3·7	4·2108	62·8	,,
MEAN	3[5]		149·6	−7·9	2·9705	15·1	
7	12	670°C	160·4	−10·3	11·3420	10·9	≤10
9	14		131·4	2·9	12·6423	9·6	,,
10	11		141·1	−1·0	10·3676	11·9	,,
11	13		118·6	−18·7	6·1227	47·0	≤20
MEAN	3[5]		144·2	−2·9	2·9205	25·1	

[1] See Figure 1.

[2] Accepted specimens have within-specimen remanence angular standard deviation ASD ≤ 10°, 20° where:

$$\mathrm{ASD} = (57{\cdot}3/R^3)((R^2-x^2)^2S_x^2+(R^2-y^2)^2S_y^2+(R^2-z^2)^2S_z^2)^{1/2}$$

$$x=(1/n)\sum_{i=1}^{n}x_i,\ S_x=(1/(n-1))\sum_{i=1}^{n}(x_i-x)^2 \quad \text{etc.;} \quad R=(x^2+y^2+z^2)^{1/2}$$

x_i, y_i, z_i are the cartesian components of the remanence vector and n is the numbers of redundant determinations ($n = 4$ for 6 spin measurement).

[3] α_{95} is the radius of the 95% confidence circle about the mean direction and R is the length of the vector resultant (Fisher 1953).

[4] ASD is the angular deviation screening criteria discussed in the text.

[5] Only homogeneously magnetised site mean directions having $\alpha_{95} \leq 20°$ are used to compute the unit mean.

Table 2 Tilt corrected tillite matrix mean remanence direction at each demagnetising temperature of 7 pilot specimens (2 specimens per site from site 7, 9, 10 and one specimen from 11)

Temperature (°C)	Declination (degrees)	Inclination (degrees)	R	R_0	α_{95} (degrees)	Intensity mean	($AM^{-1} \times 10^{-2}$) SD
20	173·2	12·1	6·7215	4·9	13·3	8·2	4·8
200	169·9	9·6	6·7779	,,	11·8	8·0	5·3
300	169·0	8·4	6·8548	,,	9·5	7·4	5·2
400	168·8	9·6	6·7953	,,	11·3	6·4	4·6
500	160·5	3·2	6·6801	,,	14·3	5·0	4·2
525	162·6	0·3	6·3831	,,	20·4	5·0	4·3
550	161·1	−2·7	6·0099	,,	26·7	5·2	4·6
565	160·7	5·6	6·6211	,,	15·6	5·3	5·1
625	150·8	0·9	6·3728	,,	20·6	4·8	9·5
640	134·2	12·6	6·1692	,,	24·0	5·0	5·2
650	131·3	1·6	6·2323	,,	23·0	7·0	9·5
660	133·8	−2·4	6·4301	,,	19·5	7·3	10·4
670	130·8	−2·7	6·5048	,,	18·2	3·4	2·7

R_0 is the 1% critical value for the length of vector resultant, R (Watson 1956). When $R > R_0$ the hypothesis that the directions are randomly directed must be rejected at the 99% level of confidence.

Table 3 Tilt corrected clast remanence data

Number of specimens	Treatment	Declination (degrees)	Inclination (degrees)	R	R_0	α_{95} (degrees)	ASD (degrees)
49	NRM	155·6	49·7	40·3418	13·6	9·5	≤10
27	,,	181·5	69·3	17·6939	10·1	20·7	>10
76	,,	160·5	56·1	57·0363	17·0	9·4	all specimens
34	20 mT	156·9	38·5	27·3148	11·3	12·4	≤10
20	,,	167·3	41·6	10·8514	8·6	31·1	>10
54	,,	159·8	39·5	38·0790	14·3	12·7	all specimens
54	550°C	155·3	14·4	38·8785	14·3	12·2	≤20°
55	650°C	140·8	16·1	25·9171	14·4	20·6	≤20°
44	670°C	141·4	2·9	19·1927	12·9	24·9	≤20°

Tarling's (1974) 'primary component' direction of 166°, −12° (α_{95}–6°). The further swing to 132°, −2° is discussed later.

The dominantly unstable remanence at matrix sites 11 and 12 has a significantly lower NRM intensity (mean $0{\cdot}44 \times 10^{-1}$ Am^{-1}; s.d. $0{\cdot}66 \times 10^{-1}$ Am^{-1}) as well as poor within-specimen (mean ASD 15·8°; s.d. 11·0°) and within-site (mean α_{95} 23·6°; s.d. 2·6°) remanence homogeneity. During step demagnetisation the remanence shows large erratic directional changes and fails to reach a stable end point (Fig. 2, spec. 11).

3.2. Tillite clasts: conglomerate test

A total of 93 specimens were cut from the 36 clast cores. Some cores drilled into smaller clasts penetrated into the matrix. Thus 14 specimens came from the matrix leaving 79 from the clasts. Of these, 25 were used for step AF and thermal demagnetisation. The remaining 54 were first AF cleaned at 20 mT and then thermally cleaned at 550, 650 and 670°C.

The post-depositional remagnetisation of the clasts is shown by the statistically nonrandom distribution of their remanence directions. Watson's (1956) test for randomness of remanence directions shows in all cases that the length of the vector resultant (R) is greater than the 1% critical value Ro (Table 3). Thus the hypothesis that the clast directions are randomly directed must be rejected at the 99% confidence level. However, both the mean directions of the NRM and of the 20 mT AF demagnetised populations are considerably steeper in inclination than the mean tillite matrix direction. Only after 550, 650 and 670°C thermal demagnetisation (Fig. 3) does the clast direction approach the matrix direction (e.g. Fig. 3a). Also, although the clast directions are statistically nonrandomly distributed, they are still relatively highly dispersed; the likely cause is incomplete remagnetisation of many specimens. In such highly dispersed populations, the mean direction often does not give a good estimate of the 'characteristic' palaeomagnetic direction because it is biased by outlier directions. The mode or 'peak of population' provides a better estimate (Van Alstine 1980), and for the clasts after the 650 and 670°C cleanings it is located at 155°, −10° (Fig. 3b, c). This mode is very similar to the mean matrix direction of 162°, 4° and to Tarling's (1974) 'primary component' direction of 166°, −12°. This similarity indicates that the tillite matrix and the siltstones were remagnetised also at approximately the same time as the clasts.

Thermal step demagnetisation of the 17 clast pilot specimens with both a small NRM ASD and a high NRM intensity were statistically examined by calculating their mean directional values at each demagnetisation step. Watson's (1956) test shows that these specimens' directions are nonrandomly distributed at >99% confidence up to 625°C. During the demagnetisation process, their direction generally moves away from the present Earth's magnetic field direction of 203°, 77° with respect to bedding towards an end point at 150°, −10°, thereby indicating the removal of a recently acquired viscous remanence component (Fig. 4).

3.3. Siltstones

Tarling (1974) showed that the siltstones carry two stable remanence components which were best isolated after 200°C thermal cleaning. For this reason, we also demagnetised our siltstone specimens at 200°C. However we found that they have poor within-specimen (mean ASD 16·4°; s.d. 5·5°) and within-site (mean α_{95} 36·8°; s.d. 17·8°) remanence homogeneity. The directions are highly dispersed (Fig. 5) and their two dominant modes of ~270°, 15° and 315°, 30° are completely different from Tarling's (1974) means of 166°, −12° and 59°, −57°. The reason for the discordance is not known, however, repeat measurements of 7 siltstone specimens shows that the directions are reproducible to only

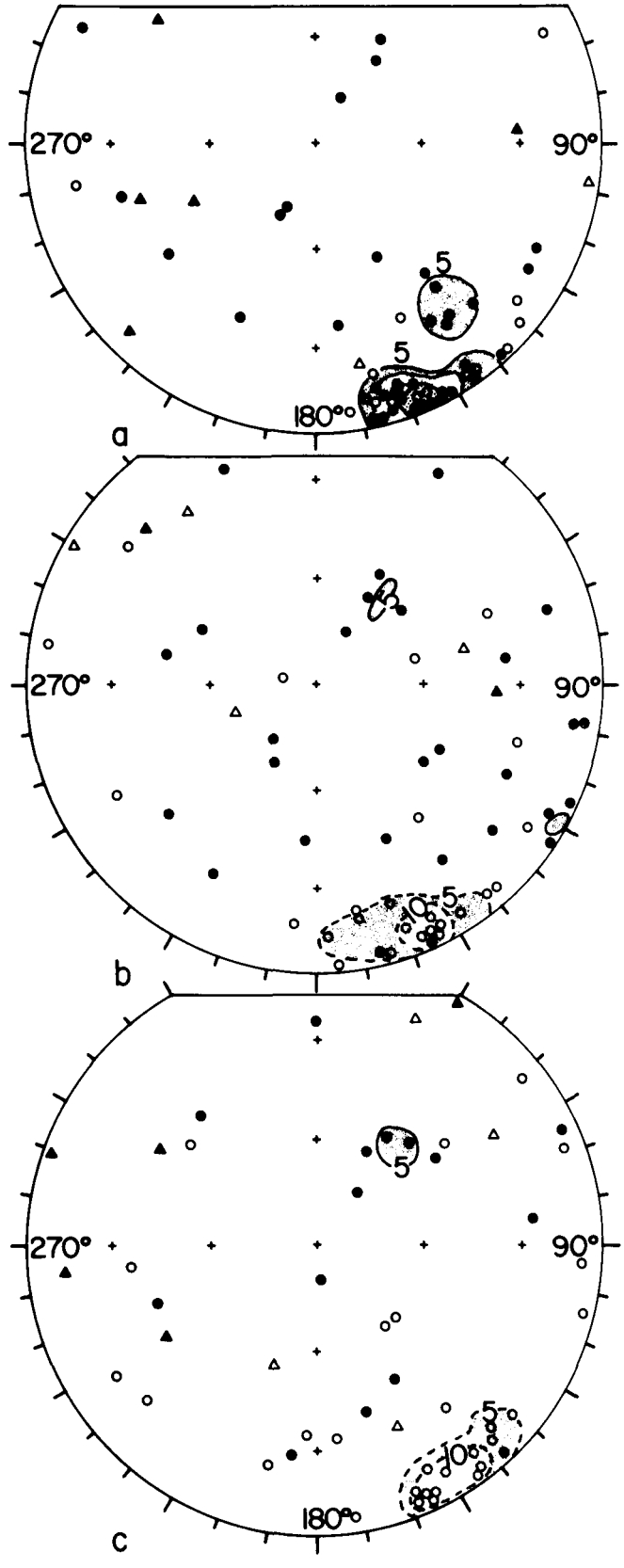


Figure 3 Stereocontour plots of all clast directions after (a) 550°C, (b) 620°C and (c) 670°C thermal demagnetisation with the circle (triangle) symbols denoting specimens with an ASD of ≤10° (>10°), the contours in % density and other conventions as in Figure 2b.

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within 18° (s.d. 8°). Also, further thermal demagnetisation of two sites at 500°C did not appreciably alter their directional distribution. In short, we are unable to determine why the siltstones obtained from the same outcrops as Tarling's exhibit different mean directions with much greater dispersion and lower stability. Given that his siltstone direction is similar to both his and our tillite matrix directions and to our tillite clast direction, we consider it to be more reliable.

4. Remanence carriers of the metamorphic remagnetization

The remagnetisation remanence carriers in the tillite matrix and in many clasts reside mostly in hematite with blocking temperatures often exceeding 670°C. These carriers are not primary hematite because even heating the tillite to 550°C for a period of 10^8 years would not have reset their magnetisation (Pullaiah *et al.* 1975). The low greenschist facies of metamorphism found in the tillite indicates that the temperature during metamorphism did not exceed ~350°C. Abouzakhm and Tarling's (1975) study also supports this contention. Thus the hematite carriers are likely to have been produced by chemical alteration at relatively low temperatures of ≤350°C to yield a highly stable chemical remanent magnetisation (CRM).

Polished section examination of three tillite matrix specimens shows only occasional unaltered small primary euhedral magnetite crystals and numerous completely or partially altered magnetite crystals that have been converted to hematite. Thin sections of the same specimens show ubiquitous hematite staining around the quartz grains and throughout the heavily altered feldspars and other matrix minerals. Polished sections of the granitic to dioritic clasts show both primary magnetite and secondary hematite from magnetite in variable proportions depending on the degree of alteration. This is consistent with the thin sections which show secondary emplacement of carbonate and veinlets of hematite especially in the biotite-rich granitic clasts. The variable nature of the alteration in the clasts provides a logical explanation for the relatively high dispersion of the clast remanence directions, i.e. they represent a mixture of randomly distributed unaltered primary clast directions grading through to nonrandomly distributed altered secondary clast directions.

5. Age of metamorphic remagnetisation

The age of the remagnetisation of the Port Askaid Tillite must be considered in relation to the apparent polar wander path (APWP) for Britain. The APWP for Cambrian to Early Devonian time is shown in Figure 6a using established poles with their ovals of 95% confidence as summarised by Morris (1976), McElhinny and Embleton (1976) and Van der Voo *et al.* (1980). Tarling's (1974) lower coercivity northerly-directed component when uncorrected for tilt (GN) does not fall on the path, but moves into the Silurian–Early Devonian polar cluster after tilt correction (GN′). This implies that this component was acquired during the Silurian upon uplift cooling or late in the orogeny, and that it is prefolding in origin. The consequence of the latter point is that the higher-coercivity southerly-directed component is also likely to be prefolding in origin. However, because its pole is located on the perimeter of the Ordovician polar cluster both before (GS) and after (GS′) tilt correction, it leads only to the conclusion that it was acquired during the Ordovician.

The results of this study (Table 4) are illustrated in Figure 6b. The matrix results (circles) both before (open) and after (solid) tilt correction are indeterminate as to the presence of a post- or prefolding remanence in that they fit into the Ordovician cluster when uncorrected or corrected for tilt.

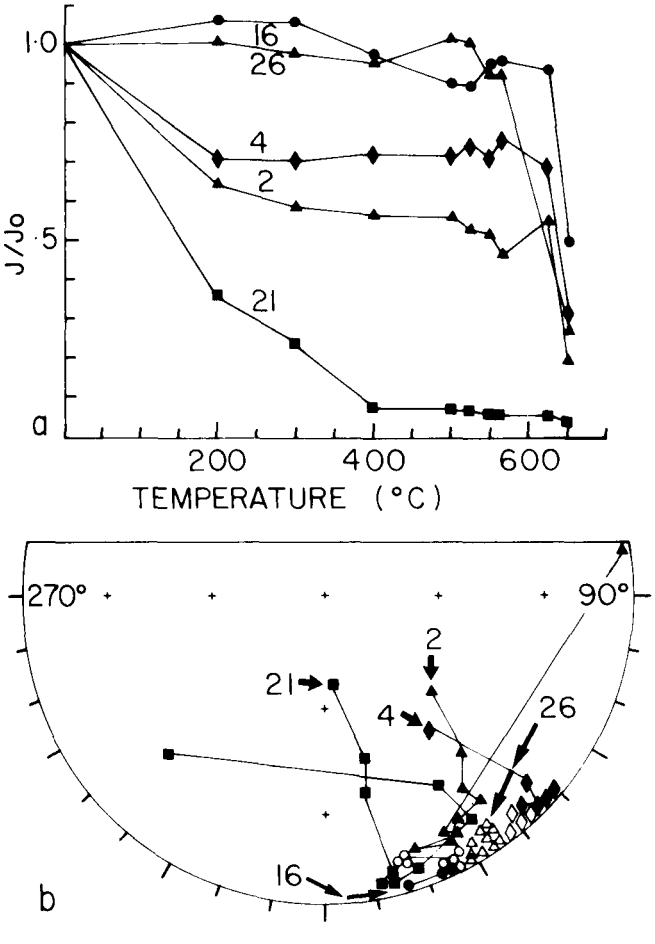


Figure 4 Thermal step demagnetisation of 5 clast pilot specimens showing (a) the remanence intensity decay as a function of the temperature in °C and (b) the remanence directions; conventions as in Figure 2.

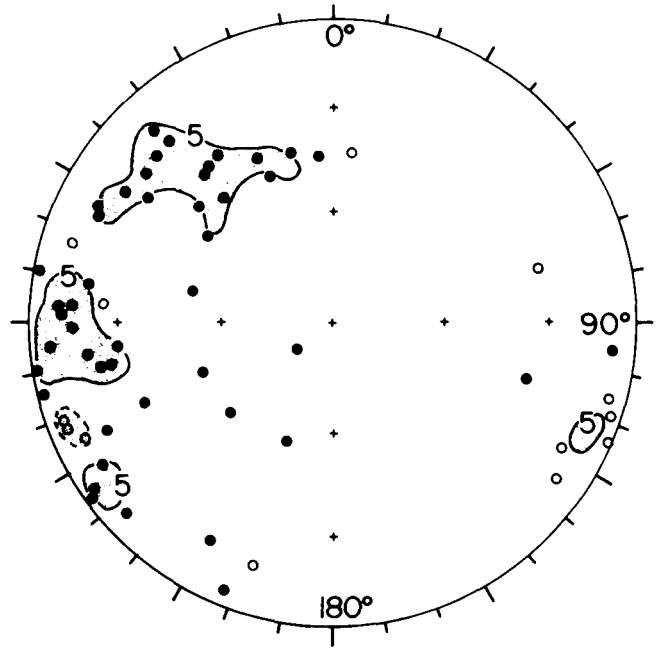


Figure 5 Stereocontour plot of the siltstone directions from 6 sites; conventions as in Figure 3.

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The lower coercivity components separated out by vector removal give a pole (VR) at the junction of the Silurian–Early Devonian and Ordovician clusters when uncorrected, and at the perimeter of the Ordovician cluster when corrected. Similarly, after AF cleaning at 20 mT and further thermal cleaning at 550°C, which gives the best defined matrix pole, the poles both before and after correction indicate an Ordovician age. Further, these matrix poles essentially duplicate Tarling's (1974) equivalent GS and GS′ poles. The matrix pole after 650°C, and the corresponding directional swing to 132°, −2° in the thermal pilot specimens, falls just outside the APWP. However this is not statistically significant when the relatively large oval of 95% confidence is taken into account. The stable matrix remanence is either pre- or postfolding and Ordovician in age.

More compelling is the evidence from the clasts. Without tectonic correction (open triangles) the NRM pole is >40° of arc away from the Ordovician cluster and >20° of arc off the APWP after 20 mT cleaning. At 550 and 650°C, the pole is close to the APWP and, given the size of the oval of 95% confidence, it is statistically the same as the Ordovician cluster. However, with the tilt correction, the poles (closed triangles) fall on the APWP path from the NRM to 550°C position, and lie insignificantly off the path after 650°C cleaning. This result indicates that the clasts were remagnetised before folding during Early Ordovician time and, by extrapolation, that the matrix remanence is also prefolding.

6. Conclusions

The upper Precambrian Port Askaid Tillite of the Dalradian Supergroup of Scotland retains a stable secondary prefolding remanence acquired during Ordovician time as shown by the conglomerate test and by the location of the tilt-corrected clast and matrix pole positions. This conclusion is supported by AF and thermal demagnetisation studies, and by polished and thin section observations which show that the remanence resides in secondary hematite derived from partially to completely altered magnetite. The results support the concept that the tillite was likely to have been deposited at mid to high palaeolatitudes during the late Precambrian. The unit then drifted into equatorial palaeolatitudes where it was chemically remagnetizated during the onset of the orogenic activity in Lower Ordovician time by regional burial metamorphism. Finally the tillite was tilted in Silurian or later time, probably as terminal orogenic event. The conglomerate test and related palaeomagnetic data demonstrate that the equatorial pole positions determined from the Port Askaid Tillite cannot be used to support the hypothesis of a worldwide glaciation during late Precambrian time.

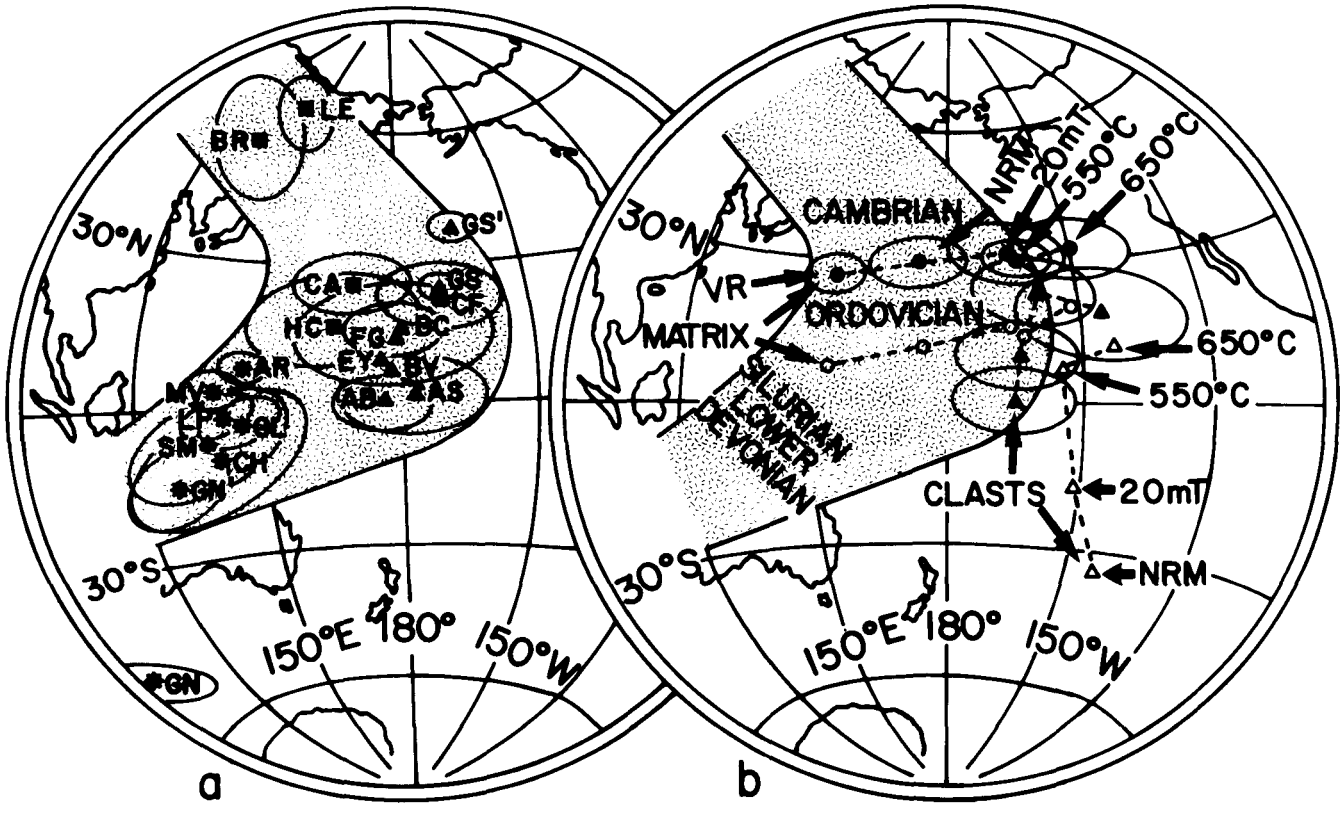


Figure 6 (a) Apparent polar wander path for Cambrian to Early Devonian time for Britain. The squares indicate Early Cambrian poles from the Leicestershire diorites (LE), Bangor red beds (BR), Caerfai series (CA), and Hartshill Quartzite (HC) used by Van der Voo *et al.* (1980). The triangles indicate Ordovician poles from the Aberdeen younger gabbros (AB), Ashgill intrusions (AS), Builth lavas (BC), Borrowdale lavas (BV), Corrock Fell complex (CF), Eycott lavas (EY), and Foyers granite (FG) used by Morris (1976). The stars indicate Silurian to Early Devonian poles from the Arrochar complex (AR), Cheviot lavas (CH), Glencoe lavas (GL), Lorne Plateau lavas (LP), Midland Valley lavas combined (MV) and the Somerset lavas (SM) used by Morris (1976); GN (GN′) and GS (GS′) are Tarling's (1974) Port Askaid sediment poles before (and after) tilt correction for the normal down and reversed equatorial components respectively; the poles are circumscribed by their oval of 95% confidence (Fisher 1953).
(b) Poles from this study plotted relative to the APWP with solid (open) symbols indicating tilt corrected (uncorrected) poles for the matrix (circles) and clasts (triangles); VR (VR′) indicates the low coercivity vector-removed pole and other lettering indicates the cleaning procedure used; the pole positions used are listed in Table 3.

https://doi.org/10.1017/S0263593300009615 Published online by Cambridge University Press

Table 4 Virtual geomagnetic poles

Number of sites (specimen)	Treatment	Uncorrected for folding VGP Long. E (degrees)	Lat. N (degrees)	dp (degrees)	dm (degrees)	Tilt corrected VGP Long. E. (degrees)	Lat. N (degrees)	dp (degrees)	dm (degrees)
26[1]	VR[2]	155·2	8·5	3·8	6·3	155·3	26·9	2·7	5·3
3	NRM, matrix	175·4	11·5	8·8	14·7	174·1	29·6	6·2	12·3
3	20 mT, ,,	193·9	16·5	8·2	15·0	193·7	32·1	6·8	13·6
4	550°C, ,,	196·6	14·0	5·8	10·3	195·6	30·0	4·6	9·2
3	650°C, ,,	207·9	19·8	8·1	15·6	210·9	32·4	7·6	15·2
3	670°C, ,,	214·4	15·9	13·8	26·3	215·7	28·2	12·6	25·1
49[1]	NRM, clasts	214·8	−34·6	14·3	16·5	195·0	0·8	8·4	12·6
34[1]	20 mT, ,,	208·8	−18·4	15·2	19·4	195·9	9·7	8·7	14·7
54[1]	550°C, ,,	205·5	5·8	8·9	14·7	201·2	23·2	6·4	12·5
55[1]	650°C, ,,	221·3	2·3	14·0	24·0	215·3	17·9	10·9	21·2
44[1]	670°C, ,,	218·2	11·7	14·3	26·7	217·4	24·4	12·5	24·9

[1] Number of specimens.
[2] Mean of 26 specimen vectors—removed by 20 mT demagnetisation.

7. Acknowledgements

This work was supported by operating grants from the Natural Science and Engineering Research Council of Canada.

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M. STUPAVSKY, D. T. A. SYMONS and C. P. GRAVENOR, Department of Geology, University of Windsor, Windsor, Ontario, Canada N9B 3P4.

MS received 11 February 1982. Accepted for publication 25 June 1982.

https://doi.org/10.1017/S0263593300009615 Published online by Cambridge University Press